

Breast Ultrasound Lesion Segmentation: an In-Browser Research Demo on BUSI with External Validation

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<https://github.com/surabhif/Breast-ultrasound-lesion-segmentation>

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Research demo — not for clinical use. Model version 1.0.0.

Abstract

Breast ultrasound helps doctors look at breast lumps, especially when mammograms are hard to read in dense tissue. This student project builds a program that outlines those lumps on ultrasound pictures and estimates how well it does at telling harmless from cancerous lumps. It was trained on one public image collection and checked on three others from different hospitals. The main quality measure is an outline-overlap score (Dice), which is high when the computer outline matches the expert outline. On images saved for testing, the browser version (11.7 MB) reaches 0.697 (95% CI [0.632, 0.766]) overall. On images that contain a lump, the outline match is 0.764. On two outside collections, outline scores stay similar (0.714 and 0.627), while telling harmless from cancerous lumps gets harder after the hospital and scanner change (0.638 / 0.721 / 0.780 versus 0.931 on the original test set). On a third outside collection, pictures without lumps remain difficult: false outlines pull the overall score down to 0.386, even though on images that contain a lump the outline match is 0.679 (false outlines on normals 320/413; BUSI normals 12/19). The write-up also studies near-identical photos, burned-in measurement marks, and whether outlines stay stable when a picture is flipped. A retrained version did better on some outside images but not consistently, so the original stays. This is an educational research demonstration, not a medical device, and must not be used for diagnosis or care decisions.

1 Introduction

Breast ultrasound is widely used as an adjunct to mammography, especially in dense breasts. Public research datasets such as BUSI [2] enable students and independent researchers to study lesion segmentation and triage-style classification, but reported numbers are easy to over-interpret when splits leak near-duplicates, images contain burned-in calipers, or models never leave the training distribution.

Student portfolio projects often stop at a training notebook. Clinically adjacent demos that run in a browser are rarer, yet they force the same engineering constraints that matter for honest reporting: a fixed preprocessing path, a quantized artifact small enough to download, and metrics that match what the user actually runs. They also make failure modes visible—empty masks on normals, brittle classification off-distribution, and optimistic BUSI scores on annotated frames—in a way a table alone does not.

This project builds an end-to-end, reproducible pipeline: download and prepare BUSI, train a compact ResNet-18 U-Net, export and quantize to ONNX, and ship a React web demo that runs inference

in the browser with onnxruntime-web. Metrics, failure galleries, and external validation live beside the demo on GitHub Pages. The goals are educational clarity and honest reporting—not clinical deployment.

Contributions. (1) An in-browser INT8 demo with training–serving parity (shared resize and morphology). (2) Grouped near-duplicate splits and a caliper/annotation audit on BUSI. (3) Pre-registered external validation of frozen model 1.0.0 on BUS-BRA and BrEaST. (4) A written MODEL_POLICY for when a retrained candidate may replace the served model. (5) Open citation/freeze scaffolding (CITATION.cff, Zenodo metadata, number-locked preprint sources) with a minted software DOI (<https://doi.org/10.5281/zenodo.23286597>).

2 Related work and positioning

U-Net-style encoders remain a default for biomedical segmentation [6]. Breast ultrasound datasets have expanded beyond BUSI to multi-center collections with patient metadata and richer clinical fields, including BUS-BRA [3], BrEaST [4], and BUS-UCLM [5]. Literature Dice numbers vary widely with splits, resolutions, and whether normals are included; we therefore treat published tables as context, not as a leaderboard to beat, and document protocol differences in the companion site page `docs/LITERATURE_COMPARISON`.

This work is deliberately scoped as a *research demo*: single student author, CPU-friendly training, browser inference, and explicit non-clinical disclaimers. We do not claim state-of-the-art segmentation. We do claim that the numbers shown next to the demo are the numbers of the served artifact, regenerated from committed JSON.

3 Methods

3.1 Datasets and splits

BUSI [2] provides breast ultrasound images with lesion masks and labels (benign, malignant, normal). The archive used here is obtained via a public Hugging Face mirror of *Dataset_BUSI_with_GT* (≈ 780 images). BUSI does not provide patient identifiers. Splits therefore keep perceptual-hash near-duplicate clusters together so near-identical frames cannot appear in both train and test. Held-out sizes: train 534, val 134, test 112. This is *not* a true patient-level split.

External sets (CC BY 4.0; scored after protocol registration, no fine-tuning): BUS-BRA [3] (1875 images, `n_patients` = 1064 in the committed JSON) and BrEaST [4] (256 images / patients). BUS-UCLM [5] is the secondary set (640 images / 38 patients after excluding Doppler/combined frames; status scored). UDIAT and BUSIS are not included (institutional licence / no-redistribution constraints).

3.2 Model and training

Architecture: ResNet-18 encoder U-Net [6] with an auxiliary benign-vs-malignant head (≈ 12.1 M parameters). Images are resized to 160×160 with a shared half-pixel-center bilinear path used in training, Python evaluation, and the browser. Training used paired flips, small rotations, scale-crop, and brightness/contrast augmentation; loss combines pos-weighted BCE and soft Dice for segmentation with BCE for classification on B/M only. Optimizer: AdamW with cosine schedule; encoder freeze for the first

few epochs; checkpoint selected on validation lesion Dice (CPU-friendly recipe; seed 42; fold 0 primary run, with a five-fold CV summary retained for diagnostics).

3.3 Post-processing and served artifact

Validation-tuned operating point (frozen for all reports): segmentation threshold 0.4, minimum connected-component area 40 pixels (4-connectivity), classification threshold 0.5. The training FP32 checkpoint is exported to ONNX and dynamically quantized to UINT8 weights for serving (11.7 MB vs 46.3 MB FP32). The live demo loads `web/public/models/v1.0.0/busi_unet.onnx`. Served INT8 metrics are computed with `onnxruntime` on CPU using the same postprocess as the browser.

3.4 Browser demo

The Vite/React TypeScript app runs `onnxruntime-web` in a Web Worker with self-hosted WASM, optional threshold sliders, sample galleries, expert-vs-model overlays, BI-RADS context pages, and Results pages backed by committed JSON. No ultrasound pixels leave the user's browser for inference; the About page states that the site uses no analytics cookies for tracking. A captioned walkthrough video and an auto-filled PDF report are shipped under `web/public/`.

3.5 External validation protocol and MODEL POLICY

Protocol text lives in `docs/EXTERNAL_VALIDATION_PROTOCOL.md` and was committed before Phase 1 scoring. External scoring uses identical preprocess/postprocess and the frozen INT8 weights. A candidate v2 may replace served v1 before the Dec 4 freeze only if it matches or beats v1 on (a) BUSI clean-subset Dice and (b) held-out / external Dice under the fairness rule below, and (c) BUSI test AUC drops by at most 0.02. Otherwise v1 remains served and v2 is published as an experiment only.

Phase 4 fairness. When v2 is trained on BUSI train plus a patient-grouped BUS-BRA train split, the BUS-BRA held-out set is *same-source held-out* for v2 (not external). We therefore score v1.0.0 INT8 on the identical held-out case IDs and compare against that baseline, not against v1's full-set BUS-BRA Dice. BrEaST remains the only truly external test. We report seed mean $\pm$ SD and paired-bootstrap 95% confidence intervals for each delta, decide promotion on the seed mean, and—if the mean passes—promote the median-performing passing seed by clean Dice (not the best).

3.6 Additional analyses

Heuristic caliper/text flags, clean-vs-flagged Dice comparisons, a short leakage ablation (grouped vs random splits), Telea inpaint retrain experiments, offline flip-based TTA uncertainty, and BrEaST measurement-agreement summaries (research sizing only) are reported on the site and in committed JSON under `results/` and `web/public/results/metrics.json`. Clinician outreach was considered early and explicitly revoked; no reviewer acknowledgements appear on the site.

4 Results

All headline numbers below are macros generated from `web/public/results/metrics.json` via `scripts/export_paper_numbers.py` (`paper/numbers.tex`). Re-export and recompile after any metrics change; `scripts/check_paper_numbers_drift.py` fails on drift.

4.1 BUSI held-out test (served INT8)

Table 1: BUSI test metrics for served INT8 (v1.0.0) vs FP32 training checkpoint.

Metric	FP32	INT8 (served)
Test Dice (all)	0.686	0.697 [0.632, 0.766]
Lesion-only Dice	0.751	0.764 [0.705, 0.821]
Test IoU	0.618	0.630
B vs M ROC-AUC	0.939	0.931
Sensitivity / Specificity	—	0.867 / 0.841
ECE	—	0.111
Normal non-empty masks	—	12/19

Quantization changes Dice only slightly (INT8 often within bootstrap noise of FP32) while shrinking the download for browser use. Classification AUC on BUSI remains high (0.931) but calibration is imperfect (ECE 0.111). The normal-class false positive rate (0.632) is a first-order usability failure for any “run on a random ultrasound” demo: empty-ground-truth frames still receive non-empty masks on 12/19 cases.

4.2 Cleaning / caliper audit

About 0.469 of BUSI images were flagged for likely calipers or burned-in text by a heuristic audit. Overall Dice on flagged test frames (0.747) exceeds clean frames (0.623); inflation of all-minus-clean Dice is 0.063. Lesion-only Dice shows a similar gap (flagged 0.799 vs clean 0.701). Readers should treat headline BUSI Dice as partly driven by easier annotated frames. Classification AUC on this stronger model is similar on clean vs flagged subsets; earlier weaker baselines showed larger gaps, so shortcut learning should be checked per recipe rather than assumed absent.

4.3 External validation (frozen INT8)

Table 2: External Dice and AUC for frozen served INT8 (no tuning).

Dataset	n	Dice	AUC
BUSI (internal INT8)	112	0.697	0.931
BUS-BRA	1875	0.714	0.638
BrEaST	256	0.627	0.721
BUS-UCLM	640	0.386 [†]	0.780

Segmentation transfers moderately (BUS-BRA Dice 0.714; BrEaST 0.627; BUS-UCLM lesion Dice 0.679). BUS-UCLM all-image Dice 0.386 is pulled down by normal false positives (320/413).

Classification AUC drops substantially off-distribution (BUS-BRA 0.638; BrEaST 0.721; BUS-UCLM 0.780), underscoring that BUSI triage scores are not portable without further work. Subgroup Dice by BI-RADS and scanner on BUS-BRA is reported in the site Results JSON for exploratory inspection; those rows were not used for threshold selection.

4.4 MODEL_POLICY outcome (inpaint retrain)

A caliper-inpaint retrain (candidate v2) slightly improved clean-subset Dice (0.626 vs v1 0.623) and did not harm AUC under the 0.02 budget, but *failed* external Dice on BUS-BRA (0.626 vs 0.714) and BrEaST (0.608 vs 0.627). The site therefore continues to serve v1.0.0. Publishing the failed swap is intentional: it shows that cleaning training images can help an internal clean subset while hurting transfer.

4.5 Uncertainty, leakage, measurements

Flip-based TTA disagreement correlates with error (Spearman $\rho \approx 0.613$); keeping the most certain 80% of cases yields mean Dice 0.778 on the BUSI test summary. This is agreement under flips—not a calibrated error probability. A short grouped-vs-random leakage ablation on this schedule did not show large positive val inflation (random-grouped ≈ -0.018); leakage risk remains conceptually important because BUSI lacks patient IDs. On BrEaST, model vs expert longest-diameter correlation (mm, exploratory) is 0.697; disagreement rate on a 20 mm threshold proxy is 0.171. Imaging size is not pathological T-stage; these figures are research diagnostics only.

5 Limitations

- **Not for clinical use.** No clinician review; not a medical device; no diagnostic claims. Site pages that mention BI-RADS or surgical sizing are educational framing only.
- BUSI lacks patient IDs; near-duplicate grouping is a proxy, not a guarantee against leakage.
- High false-positive rate on normals (12/19).
- External classification AUC is much lower than internal BUSI AUC.
- Resolution is limited to 160^2 for CPU/browser practicality.
- BUS-UCLM all-image Dice is low (0.386) largely from normal false positives (320/413); lesion Dice 0.679 is the fairer transfer figure.
- Caliper detector is heuristic, not OCR; literature comparisons depend on heterogeneous protocols across papers.
- Single primary training seed for the served model; CV folds are diagnostic, not an ensemble deployed in the browser.

6 Ethics, licensing, and credit

Code in this repository is released under the MIT license. BUSI redistribution terms remain those of the original dataset providers; only a minimal cited sample ships in the web tree. External demo samples use CC BY sources with attribution. Personal information on the site is limited to the author’s name and “high-school senior, Georgia”.

7 AI-use disclosure

Project by Surabhi Fadnavis. The code, analysis, text, and video were produced with AI tools (Cursor).

8 Reproducibility and availability

Code, metrics JSON, ONNX weights, and the GitHub Pages demo are in <https://github.com/surabhif/Breast-ultrasound-lesion-segmentation>. Citation metadata: `CITATION.cff`; Zenodo deposit metadata: `.zenodo.json`. Software DOI: <https://doi.org/10.5281/zenodo.23286597> (cite as [1]; agents must not create tags or releases). Paper numbers:

```
python scripts/export_paper_numbers.py
cd paper && pdflatex main && pdflatex main
python scripts/check_paper_numbers_drift.py
```

Venue options (JEI vs arXiv vs site-only) are compared in `docs/SUBMISSION_GUIDE.md`; nothing in this preprint package constitutes a submission.

Acknowledgments

No clinician reviewers. Training used public datasets cited above. Hosting is GitHub Pages.

References

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